

$$\begin{array}{c}
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\text{UF}[UA, UB] \xrightarrow{\text{UF}[U\epsilon_A, \text{id}]} \text{UF}[\text{UF}UA, UB] \xrightarrow{s_{UA, UB}} [UA, \text{UF}UB] \xrightarrow{[\text{id}, U\epsilon_B]} [UA, UB] \\
\downarrow \text{UF}[\text{id}, \eta_{UB}] \quad \equiv \quad \downarrow \text{UF}[\text{id}, \eta_{UB}] \quad \downarrow \text{def } s \quad \uparrow [\eta_{UA}, \text{id}] \\
\text{UF}[UA, \text{UF}UB] \rightarrow \text{UF}[\text{UF}UA, \text{UF}UB] \quad [\text{UF}UA, \text{UF}UB] \quad [\text{UF}UA, \text{UF}UB] \quad [\text{UF}UA, \text{UF}UB] \\
\downarrow \text{UF}[U\epsilon_A, \text{id}] \quad \downarrow \text{UF}\varphi_{FUA, FUB}^{-1} \quad \uparrow \varphi_{FUA, FUB} \\
\text{UF}\varphi_{A, B}^{-1} \quad \text{nat } \varphi_{+adj} \quad \text{UF}\varphi_{A, FUB}^{-1} \quad \text{nat } \varphi^{-1} \quad \text{UF}U[FUA, FUB] \rightarrow U[FUA, FUB] \quad \text{nat } \varphi_{+adj} \quad \uparrow \varphi_{A, FUB} \\
\downarrow \text{UF}U[\epsilon_A, \text{id}] \quad \text{UF}U[\epsilon_A, \text{id}] \quad \text{U}\epsilon_{[FUA, FUB]} \quad \text{nat } \epsilon \quad \leftarrow U[\epsilon_A, \text{id}] \\
\text{UF}U[A, FUB] \xrightarrow{\text{U}\epsilon_{[A, FUB]}} U[A, FUB] \quad \text{nat } \epsilon \quad \leftarrow U[A, \epsilon_B] \\
\text{UF}U[A, B] \xrightarrow{\text{U}\epsilon_{[A, B]}} U[A, B] \xrightarrow{\varphi_{A, B}} [UA, UB]
\end{array}
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